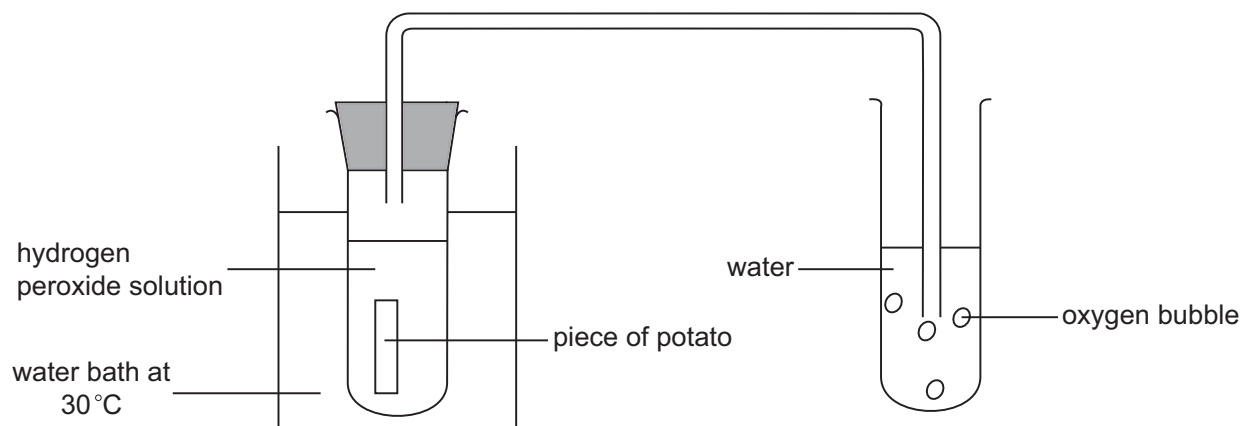


- 58** A student carried out an experiment to investigate how temperature affects the rate of activity of an enzyme found in potatoes. This enzyme breaks down the substrate hydrogen peroxide to produce water and oxygen.

The apparatus was set up as shown in the diagram and the experiment carried out at 30 °C.



The student counted the number of bubbles of oxygen produced in one minute.

The experiment was repeated at 40 °C, 50 °C, 60 °C, and 70 °C.

The table shows the results collected by the student.

<i>temperature / °C</i>	<i>number of bubbles of oxygen gas produced in one minute</i>
30	32
40	27
50	23
60	8
70	2

A group of students were given these results and asked to calculate the percentage decrease in the number of bubbles produced between 30 °C and 60 °C. Each student was also asked to write a conclusion to explain the results. These are shown in the table below.

Which student correctly calculated the percentage decrease in the number of bubbles between 30 °C and 60 °C and wrote an appropriate conclusion which explains the reason for this decrease?

	<i>percentage decrease in the number of bubbles between 30 °C and 60 °C</i>	<i>conclusion</i>
A	20	the high temperature causes the bonds maintaining the structure of the enzyme to break
B	20	the temperature of the environment is too hot and the enzyme dies
C	75	the high temperature causes the bonds maintaining the structure of the enzyme to break
D	75	the temperature of the environment is too hot and the enzyme dies
E	300	the high temperature causes the bonds maintaining the structure of the enzyme to break
F	300	the temperature of the environment is too hot and the enzyme dies